Recommendation of Agriculture Crop Based on Productivity and Season

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ABSTRACT:

Agriculture faces challenges such as unpredictable weather, soil degradation, and traditional crop selection methods that lower productivity. This project presents a Machine Learning–based Crop Recommendation System that suggests the most suitable crops by analyzing soil nutrients (N, P, K), pH, temperature, humidity, and rainfall.

Several algorithms were tested, and the Random Forest model achieved the best accuracy of about 97%, delivering reliable, data-driven recommendations. This system supports precision agriculture, helping farmers improve yield, conserve resources, and practice sustainable farming.

Keywords: Crop Recommendation, Machine Learning, Random Forest, Precision Agriculture, Sustainable Farming.

Introduction

- Agriculture is an important part of every country's economy because it provides food, jobs, and raw materials. But today, farming faces many problems like unpredictable weather, irregular rainfall, and rising temperatures, which reduce crop production. Good farming depends on factors such as sunlight, soil type, rainfall, temperature, humidity, and the proper use of fertilizers and pesticides. So, choosing the right crop and knowing when to plant and harvest are very important.

- India has four main seasons—**winter (December–March), summer (April–June), monsoon (July–September), and post-monsoon or autumn (October–November)**. Each season affects which crops can grow best.

- With the fast growth of technology, farming is changing. Information Technology (IT) and data analysis help farmers make better decisions. By using tools like machine learning and weather data, farmers can get timely advice on which crops to grow and how to manage their fields. This helps increase production, saves resources, and supports sustainable farming for the future.

Problem Statement

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Farmers often do not have data-based tools to help them choose the right crops for each season. As a result, they face lower yields and financial losses.

Common Problems

- Unstable crop production because of changing weather and climate.
- Reliance on old traditional methods that no longer meet modern needs.
- Few crop recommendation systems that consider seasonal changes.
- Lack of easy-to-use tools for managing crops.
- Existing solutions mainly focus on soil conditions and ignore seasonal factors.

Aim & Objectives

Aim:
Build a web-based crop recommendation system

Objectives:
- User-friendly interface (Django)
- Apply ML models (RF, DT, SVM, KNN, NB)
- Data visualization (graphs/charts)
- Support CSV/Excel input
- Enable real-time predictions

End-to-End Flow

Data Collection

→ Data Cleaning
→ Feature Selection
→ Model Training
→ Evaluation
→ Crop Recommendation Output

System Architecture

User Input

→ Django Frontend
→ Python ML Backend
→ Trained Model
→ Prediction
→ Result Display

Core Logic

- Input: Soil (N, P, K, pH), temperature, humidity, rainfall
- Processed using ML algorithms
- Predicts best crop with highest productivity

Data Handling

- Handled missing values
- Normalization applied
- Encoded categorical data
- Feature selection for accuracy

Models & Performance

Used models:

- Random Forest
- Decision Tree
- SVM
- KNN
- Naïve Bayes
- Best Model: **Random Forest (~97%)**

Debugging & Edge Cases

- Exception handling used
- Logging and validation applied

Handled:

- Missing inputs
- Wrong data
- Extreme weather conditions

Edge Cases

- Missing input values
- Incorrect user data
- Extreme weather conditions
- Unseen/new data

Libraries Used

- Pandas
- NumPy
- Scikit-learn
- Django
- Matplotlib

AI Usage

- Used supervised machine learning
- Analyzed patterns in agricultural data
- Predicted best crop based on inputs

Conclusion

The project develops a machine learning–based crop recommendation system that helps farmers choose suitable crops using soil, weather, and seasonal data. It improves decision-making, increases productivity, and reduces losses. With future integration of real-time data and advanced models, the system can become more accurate and practical for real-world agricultural use.

Improvements

- Real-time weather API integration
- IoT-based soil sensors
- Mobile application
- Use deep learning models

THANK YOU